DEM Comparator

QGIS Plugin for Multi-Source DEM Comparison

Download, align and compare 11 global DEMs in one click

QGIS · DEM · DSM · DTM · Copernicus GLO-30 · FABDEM · NASADEM · AW3D30 · SRTM · OpenTopography · HuggingFace

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1. Purpose and Capabilities

DEM Comparator is a QGIS plugin that lets you, in a single request, download several global Digital Elevation Models (DEM) for the same area of interest, align them to a common grid, and compare them pairwise with detailed statistics and an auto-generated HTML report. The plugin is designed for professionals who need to choose the right DEM for a specific task — hydrological modelling, forest inventory, urban canopy height estimation — and who need to understand exactly how the available datasets differ in their region of work.

Eleven DEM sources are supported, grouped into three access categories. Four sources are accessible through Microsoft Planetary Computer (PC) without an API key: Copernicus GLO-30, NASADEM and ALOS AW3D30. One source — FABDEM V1-2 (Forest And Buildings removed DEM) — is accessible directly through a HuggingFace bucket, also without an API key. Six more sources are accessible through OpenTopography (OT) and require a free registration: SRTM GL1/GL3, ALOS-OT, COP90-OT, EU-DEM and GEDI L3. All three backends share the same workflow: the user draws a bbox on the map, selects the sources and clicks "Download and compare" — the rest happens automatically.

Key use cases include: selecting the optimal DEM for a region (comparing RMSE and systematic bias across pairs), estimating vegetation and building height by subtracting a DTM from a DSM (e.g. GLO-30 − FABDEM yields a forest canopy height map), validating proprietary data against reference global DEMs, and calibrating hydrological models based on a comparison of several sources. The output of each run is a set of aligned rasters, pairwise difference rasters, and an HTML report with statistics (mean bias, MAE, RMSE, std, min, max) for every pair of sources.

The plugin is designed to keep the entry barrier as low as possible: four out of eleven sources require no registration whatsoever, FABDEM works on the same terms as PC sources, and OT sources only require a free academic registration. All rasters are loaded via the standard GDAL `/vsicurl/` mechanism, which enables streaming reads without downloading entire tiles — particularly important for large areas where only a fragment of each tile is needed.

2. Data Sources — Comparison Table

The table below lists all eleven supported DEM sources with their key characteristics. **DSM** (Digital Surface Model) is a surface model: radar sees rooftops and tree crowns, i.e. the first solid surface from above. **DTM** (Digital Terrain Model) is a bare-earth model: only the ground, without buildings or vegetation. The difference DSM − DTM is approximately equal to the height of objects above the ground (forest, buildings). This distinction is key to choosing the right source for a task.

ID	Name	Type	Res. (m)	Coverage	Host	API	License
glo30	Copernicus GLO-30	DSM	30	Global 60°S–80°N	Planetary Computer	—	Copernicus License
nasadem	NASADEM	DSM	30	Global 60°S–60°N	Planetary Computer	—	Public Domain
aw3d30	ALOS AW3D30	DSM	30	Global 82°S–82°N	Planetary Computer	—	JAXA EORC Terms
fabdem	FABDEM V1-2	DTM	30	Global 60°S–80°N	HuggingFace bucket	—	CC BY-NC-SA 4.0
SRTM_GL1	SRTM GL1	DSM	30	60°S–60°N	OpenTopography	OT	Public Domain
SRTM_GL3	SRTM GL3	DSM	90	60°S–60°N	OpenTopography	OT	Public Domain
ALOS_AW3D30_OT	ALOS AW3D30 (OT)	DSM	30	82°S–82°N	OpenTopography	OT	JAXA EORC Terms

ID	Name	Type	Res. (m)	Coverage	Host	API	License
COP90_OT	Copernicus GLO-90 (OT)	DSM	90	Global	OpenTopography	OT	Copernicus License
EU_DTM	EU-DEM	DTM	30	Continental Europe	OpenTopography	OT	EEA CC-BY 4.0
GEDI_L3	GEDI L3	DTM	1000	Vegetated regions	OpenTopography	OT	Public Domain
srtm	SRTM v3 (legacy)	DSM	30	60°S–60°N	Planetary Computer	—	Public Domain

Coverage is given for latitude. All global sources cover longitude from −180° to +180°. SRTM is limited to ±60°N, EU-DEM covers only continental Europe, GEDI L3 covers only vegetated regions. **API**: "—" means no key required; "OT" means a free OpenTopography API key is required.

3. Backends and Data Access

3.1 Microsoft Planetary Computer (PC)

Microsoft Planetary Computer hosts several global DEMs as STAC collections: **cop-dem-glo-30** (Copernicus GLO-30), **nasadem**, **alos-dem** (AW3D30) and **srtm-gl-1** (SRTM GL1). Access is free, no API key required, but URLs must be signed via the `/api/sas/v1/sign` endpoint. The token is valid for ~60 minutes and is cached by the plugin in a `pc_tokens.json` file in the QGIS profile. STAC search is performed through the standard `/api/stac/v1/search` endpoint with a bbox filter; for each tile found, a signed URL is constructed and passed to GDAL via `/vsicurl/`.

```
Endpoints: https://planetarycomputer.microsoft.com/api/stac/v1 (STAC),
https://planetarycomputer.microsoft.com/api/sas/v1/sign (URL signing).
```

3.2 HuggingFace bucket (FABDEM)

FABDEM V1-2 is hosted on neither PC nor OT. The original distributor (University of Bristol, `data.bris.ac.uk`) publishes the data as 10°×10° ZIP archives with plain GeoTIFFs inside — inconvenient for windowed reads. The Italian non-profit Fondazione LINKS (AI, Data & Space) reprocessed the data into Cloud-Optimized GeoTIFF (COG) format and published it on HuggingFace: dataset `links-ads/fabdem-v12` with STAC metadata and bucket `links-ads/fabdem` with 19,011 COG tiles (470 GB).

Access is direct, no authentication, range-readable, CORS open. Tiles are float32, 3600×3600 px (1°×1°), EPSG:4326 + EGM2008, DEFLATE-compressed, with internal overview pyramids (2/4/8). The plugin builds a list of candidate tiles from the bbox, HEAD-filters missing (ocean) tiles and passes the surviving URLs into the same `build_vrt_for_items` + `warp_to_output` pipeline used for PC sources.

```
URL pattern:
https://huggingface.co/buckets/links-ads/fabdem/resolve/tiles/<GROUP>/<TILE>_FABDEM_V1-2.tif
Example: …/tiles/N40E000-N50E010_FABDEM_V1-2/N44E007_FABDEM_V1-2.tif
```

3.3 OpenTopography (OT)

OpenTopography is an academic service providing access to several global and regional DEMs via a REST API. A free API key is required, obtainable after registration on the portal. The plugin uses the `/globaldem` endpoint, which returns a bbox-clipped GeoTIFF directly — no VRT mosaic needed. OT sources are only enabled if the user has entered an API key in the plugin dialog; without a key, only PC and HF sources are available.

```
Endpoint: https://portal.opentopography.org/API/globaldem
Free-tier limits: 50 requests / 24 h (non-academic), 200 / 24 h (academic), 450 km² per
request at 30 m resolution.
```

4. OpenTopography API Key and Installation

4.1 Obtaining a free OT API key

If you plan to use SRTM GL1/GL3, ALOS-OT, COP90-OT, EU-DEM or GEDI L3, a free OpenTopography API key is required. Registration takes 5–10 minutes and involves no payment. First, register on the portal at `portal.opentopography.org/newUser` (name, surname, affiliation, password). After registering, sign in to your dashboard at `portal.opentopography.org/myopentopo` and click "Request API key", or go directly to `portal.opentopography.org/requestService?service=api`. Fill in a short form describing your use case (one sentence is enough, e.g. "DEM comparison in QGIS for forestry analysis"). The key is sent by email within a few minutes; for academic users, usually instantly.

Free-tier limits: 50 requests per day for non-academic users and 200 requests per day for academic users. A single request covers up to 450 km² at 30 m resolution — enough for most regional tasks: one request covers an area of roughly 21×21 km. For larger areas, the plugin does not auto-split the bbox — you need to draw a bbox smaller than the limit. The API key field in the plugin dialog has a "Get a key..." button that opens the request form in a browser; the field tooltip contains step-by-step instructions. The key can be persisted between QGIS sessions by checking "Remember".

4.2 Plugin installation

The plugin is distributed as a ZIP archive with the standard QGIS plugin structure: a top-level `dem_comparator/` folder containing `metadata.txt`, `__init__.py`, `plugin.py` and subfolders `sources/`, `ui/`, `core/`, `icons/`. To install, open **Plugins → Manage and Install Plugins → Install from ZIP** in QGIS, select the `dem_comparator_fixed.zip` archive and confirm. QGIS will automatically unpack the archive into `~/.local/share/QGIS/QGIS3/profiles/default/python/plugins/dem_comparator/` and register the plugin.

After installation, the DEM Comparator toolbar appears under **Raster → DEM Comparator**. Minimum QGIS version: 3.34 LTS. No external Python dependencies are required: the plugin uses only the Python standard library and GDAL, which ships with QGIS. If the `requests` library is available in the system, it is used for HTTP requests; otherwise, the plugin falls back to `urllib` from the standard library.

5. Licenses and Attribution

Each DEM source has its own license that must be respected when using the data in publications, reports or commercial products. In particular: **FABDEM is distributed under CC BY-NC-SA 4.0** (NonCommercial-ShareAlike) — this means the data cannot be used for commercial purposes without separate permission from the rights holder (Fathom Global, contact: `fabdem@fathom.global`). Any use of FABDEM requires citation of Hawker et al. (2022). The other sources are either Public Domain (NASADEM, SRTM, GEDI) or have permissive licenses that allow commercial use with attribution.

Source	License	Commercial use	Citation
Copernicus GLO-30	Copernicus License (free)	Yes	© DLR e.V. 2010-2014, © Airbus DS 2014-2018, Copernicus ESA

Source	License	Commercial use	Citation
NASADEM	Public Domain (NASA)	Yes	NASA JPL (2020). NASADEM
ALOS AW3D30	JAXA EORC Terms	Limited	JAXA (2021). ALOS World 3D-30m
SRTM GL1/GL3	Public Domain (NASA/USGS)	Yes	USGS (2014). SRTM v3
FABDEM V1-2	CC BY-NC-SA 4.0	NO (non-commercial)	Hawker et al. (2022), Environ. Res. Lett. 17(2) 024016
EU-DEM	EEA CC-BY 4.0	Yes	European Environment Agency (2016)
GEDI L3	Public Domain (NASA)	Yes	Dubayah et al. (2020). GEDI L3

5.1 Links to original sources

FABDEM V1-2 (University of Bristol, original):
`https://data.bris.ac.uk/data/dataset/s5hqmjcdj8yo2ibzi9b4ew3sn` · DOI: 10.5523/bris.s5hqmjcdj8yo2ibzi9b4ew3sn

FABDEM V1-2 (HuggingFace mirror, COG + STAC):
`https://huggingface.co/datasets/links-ads/fabdem-v12` · bucket:
`https://huggingface.co/buckets/links-ads/fabdem`

Microsoft Planetary Computer (GLO-30, NASADEM, AW3D30):
`https://planetarycomputer.microsoft.com` · STAC:
`https://planetarycomputer.microsoft.com/api/stac/v1`

OpenTopography (SRTM, ALOS, COP90, EU-DEM, GEDI): `https://opentopography.org` · API:
`https://portal.opentopography.org/API`

5.2 FABDEM citation (mandatory)

Hawker, L., Uhe, P., Paulo, L., Sosa, J., Savage, J., Sampson, C., & Neal, J. (2022). A 30 m global map of elevation with forests and buildings removed. *Environmental Research Letters*, 17(2), 024016. `https://doi.org/10.1088/1748-9326/ac4d4f`

Any public use of results derived from FABDEM (a scientific paper, a report, a map, a website) must include the above reference. For commercial use, a separate permission must be obtained from Fathom Global (`fabdem@fathom.global`).

6. How Comparison Works

The plugin's full pipeline runs in five stages. First, the orchestrator (`core/orchestrator.py`) filters the selected sources by coverage and discards those that cannot serve the requested bbox (e.g. SRTM for territory above 60°N). Then, for each source, the `download(extent, output_path)` method is called, which fetches the rasters and saves them into a `raw/<source_id>/` subfolder. At the third stage, all rasters are regridded to a common grid (chosen resolution, EPSG:4326, a chosen resampling algorithm — bilinear, cubic or nearest) and saved into `aligned/`. The fourth stage performs pairwise comparison of all aligned rasters, computes statistics and saves difference rasters to `diffs/`. The final stage generates the HTML report.

6.1 Raster alignment (aligned/ folder)

After download, rasters in `raw/` are in their "native form" — each source with its own geometry, resolution and coordinate system. GLO-30, AW3D30 and SRTM arrive as a set of 1°×1° tiles in EPSG:4326 at 30 m. NASADEM is also 1°×1° tiles, but heights are referenced to the WGS84 ellipsoid via EGM96, while GLO-30 and AW3D30 use the EGM2008 geoid. FABDEM is 1°×1° tiles in EPSG:4326 referenced to EGM2008, float32. OT sources return a single bbox-clipped GeoTIFF at their native resolution (30 or 90 m). EU-DEM arrives in EPSG:3035 (Europe equal-area) at 25 m, and GEDI L3 is rasterized points at ~1 km. If such rasters were subtracted directly, pixels would not match in coordinates, resolution or vertical datum — the comparison would be meaningless.

The alignment stage (`core/align.py`) brings all rasters to a common reference. First, a common extent is computed — the bbox drawn by the user (or the union of available extents if the bbox exceeds the coverage of any single source). Then a common resolution is set — 30 m by default, but switchable to 90 m when comparing with COP90 or SRTM_GL3, or to 1 km for GEDI. All rasters are divided into pixels with the same bounds and grid step `(xmin, ymin, xmax, ymax, pixel_size_x, pixel_size_y)`, so the pixel counts along X and Y match across all sources. All rasters are reprojected to EPSG:4326 (WGS84, the QGIS default). The vertical datum is unified to WGS84 (ellipsoidal): FABDEM and NASADEM are additionally transformed from EGM2008/EGM96 using GDAL `gdalwarp` with the NOAA EGM lookup grid. The resampling algorithm is uniform across sources — bilinear (default for continuous terrain), switchable to cubic (sharper, better for detailed relief) or nearest (for categorical data — not used for DEMs). The Nodata value is also unified to −9999, so that NaN pixels are handled correctly during subtraction and do not produce artifacts such as 0 − 0 = 0 on empty areas.

Technically, alignment is implemented via GDAL VRT + `gdal.Warp`: first, each source's tiles are combined into a virtual mosaic (VRT) without merging data, then the VRT is passed to `gdal.Warp`, which performs reprojection, resampling and bbox clipping in a single pass. The result is saved to `aligned/<source_id>.tif` — one file per source — and all of them **match pixel-by-pixel**: pixel `[i, j]` in `glo30.tif` corresponds to the same patch of ground as pixel `[i, j]` in `fabdem.tif`. Thanks to this, subtraction becomes a simple operation `pixel_a[i,j] − pixel_b[i,j]` with no additional georeferencing; without the alignment stage, all subsequent analysis would be impossible.

6.2 Comparison metrics

For each pair of sources (A, B), a difference raster A − B is computed, along with the following statistics over valid pixels (i.e. pixels where both rasters have non-Nodata values): **Mean bias** — the average offset (if A is systematically higher than B, the bias is positive); **MAE** (Mean Absolute Error) — the average absolute deviation, robust to sign; **RMSE** (Root Mean Square Error) — the root of the mean squared error, sensitive to outliers; **Std** — the standard deviation of the differences; **Min/Max** — extreme difference values, useful for diagnosing tile-boundary artifacts. The number of compared pairs for N sources is N×(N−1)/2 — e.g. for 4 sources, 6 pairs; for 6 sources, 15 pairs.

6.3 Interpretation example

On a test site in Sakhalin (~11×16 km, taiga), the four-source comparison gave the following key results. The pair GLO-30 − SRTM showed a mean bias of +1.0 m and an RMSE of 4.1 m — both sensors are radar-based and see the same surface (tree canopies); the small difference is explained by different acquisition dates and processing algorithms. The pair NASADEM − SRTM gave an RMSE of just 1.5 m — NASADEM is a reprocessed SRTM, so the two are nearly identical. The pair GLO-30 − FABDEM showed a mean bias of +8.9 m — this is precisely the average forest canopy height over the site: GLO-30 (DSM) sees the crowns, FABDEM (DTM) sees the ground beneath the forest. The minimum RMSE of 1.5 m (NASADEM − SRTM) is automatically flagged by the plugin in the report as the "most consistent pair" — a useful hint for choosing a reference source.

7. Links and Contacts

7.1 Data sources

Resource	URL
Microsoft Planetary Computer (STAC)	https://planetarycomputer.microsoft.com/api/stac/v1
Microsoft Planetary Computer (URL signing)	https://planetarycomputer.microsoft.com/api/sas/v1/sign
OpenTopography API (globaldem)	https://portal.opentopography.org/API/globaldem
OpenTopography registration	https://portal.opentopography.org/newUser
OpenTopography API key request	https://portal.opentopography.org/requestService?service=api
FABDEM (original, Bristol)	https://data.bris.ac.uk/data/dataset/s5hqmjcdj8yo2ibzi9b4ew3sn
FABDEM (HF dataset, STAC)	https://huggingface.co/datasets/links-ads/fabdem-v12
FABDEM (HF bucket, COG tiles)	https://huggingface.co/buckets/links-ads/fabdem
Copernicus DEM (ESA)	https://spacedata.copernicus.eu/web/cscda/data-access/section-3
NASADEM (NASA)	https://earthdata.nasa.gov/esds/competitive-programs/measures/nasadem
ALOS World 3D (JAXA)	https://www.eorc.jaxa.jp/ALOS/en/aw3d30/
EU-DEM (EEA)	https://land.copernicus.eu/imagery-in-situ/eu-dem
GEDI L3 (NASA)	https://gedi.umd.edu/data/products/

7.2 FABDEM citation (Hawker et al., 2022)

Hawker, L., Uhe, P., Paulo, L., Sosa, J., Savage, J., Sampson, C., & Neal, J. (2022). A 30 m global map of elevation with forests and buildings removed. *Environmental Research Letters*, 17(2), 024016.

```
DOI: https://doi.org/10.1088/1748-9326/ac4d4f
```

7.3 Contacts

Plugin support and maintenance: Nadezhda Vladimirova (`nadiopt@yandex.ru`). For installation, usage, bug reports or proposals of new data sources, please contact this address.

For commercial use of FABDEM: `fabdem@fathom.global` (Fathom Global, UK). For questions about access via the HuggingFace mirror: dataset `links-ads/fabdem-v12` (Fondazione LINKS — AI, Data & Space, Italy). For PC data questions: `https://planetarycomputer.microsoft.com/contact`. For OT API questions: `info@opentopography.org`.